Module 2 M&A valuation with multiples/comparables

1. You are provided with the following information about a firm:

- The number of shares outstanding is 20 million.
- The current share price is $5 per share.
- The total value of the firm's debt is $50 million.
- The cash and cash equivalents of the firm are $5 million.
- The firm does not have any non-controlling interests.
- The net income of the firm is $10 million.
- The firm's EBITDA is $20 million.

What is the P/E ratio of the firm?

There are two ways that you can calculate the P/E:

(1) at per share level:

P/E= price per share / Earnings per share (EPS)

EPS = Net Income (NI)/ Shares outstanding = $ 10 million/20 million= $0.5 per share.

P/E = $5 / $0.5 =10.

(2) at firm level:

P/E = Mkt cap or total market value of equity divided by NI

Mkt cap = share price x shares outstanding = $5 x 20 million = $100 million

P/E = $100 million/ $10 million = 10.

2. You are provided with the following information about a firm:

- The number of shares outstanding is 20 million.
- The current share price is $5 per share.
- The total value of the firm's debt is $50 million.
- The cash and cash equivalents of the firm are $5 million.
- The firm does not have any non-controlling interests.
- The firm's EBITDA is $20 million.

What is the enterprise value of the firm?

Step 1: Mkt value of equity

Mkt value of equity is also called "mkt cap": = shares outstanding x share price = 20 mln x $5 = $100 mln.

Step 2: EV

EV = Mkt value of equity + debt + noncontrolling interest (also called minority interest) - cash & cash equivalents = \$100 mln + \$50 mln + 0 - \$5 mln = \$145 mln.

3. You are provided with the following information about a firm:

- The number of shares outstanding is 20 million.
- The current share price is \$5 per share.
- The total value of the firm's debt is \$50 million.
- The cash and cash equivalents of the firm are \$5 million.
- The firm does not have any non-controlling interests.
- The firm's EBITDA is \$20 million.

What is the EV/EBITDA of the firm?

Step 1: Mkt value of equity

Mkt value of equity is also called "mkt cap": = shares outstanding x share price = 20 mln x \$5 = \$100 mln.

Step 2: EV

EV = Mkt value of equity + debt + noncontrolling interest (also called minority interest) - cash & cash equivalents = \$100 mln + \$50 mln + 0 - \$5 mln = \$145 mln.

Step 3: EV/EBITDA = 145/20=7.25

4. You are tasked with valuing Waypoint Inc, which has a net income of \$20 million and 10 million shares outstanding. You identify the following three comparable firms that have similar risk and growth as Waypoint:

Comparable Firms	P/E
A-Cap Ltd	14
Zimi Ltd	15
APN Corp.	16

What should be the estimated share value of Waypoint if you use the comparables/multiples approach?

Step 1: Average multiple of the comparable firm:

= (14+15+16)/3=15

Step 2: Earnings per share of the target firm:

= Net Income/shares outstanding = \$20 million/10 million = \$2 per share

Step 3: Estimate the share value

= average multiple from step 1 x the target's EPS from step 2

= 15 x $2 = $30 per share.

5. You are tasked with valuing Waypoint Inc, which has an EBITDA of $30 million and 10 million shares outstanding. You identify the following three comparable firms that have similar risk and growth as Waypoint:

Comparable Firms	EV/EBITDA
A-Cap Ltd	19
Zimi Ltd	20
APN Corp.	21

What should be the estimated enterprise value (EV) of Waypoint if you use the comparables/multiples approach?

Step 1: Average multiple of the comparable firm:

= (19+20+21)/3=20

Step 3: Estimate the EV

= Average multiple from step 1 x the target's EBITDA= 20 x $30 mln = $600 mln.

6. You are tasked with valuing Waypoint Inc, which has an EBITDA of $30 million and 10 million shares outstanding. Additionally, Waypoint has $100 million of cash and cash equivalent, $100 million of debt , and zero noncontrolling interests. You identify the following three comparable firms that have similar risk and growth as Waypoint:

Comparable Firms	EV/EBITDA
A-Cap Ltd	19
Zimi Ltd	20
APN Corp.	21

What should be the estimated share value of Waypoint if you use the comparables/multiples approach?

Step 1: Average multiple of the comparable firm:

= (19+20+21)/3=20

Step 3: Estimate the enterprise value (EV)

= average multiple from step 1 x the target's EBITDA= 20 x $30 mln = $600 mln.

Step 4: Estimate the market value of equity or market cap

Note: EV = mkt value of equity + debt + noncontroling interest - cash & cash equivalent

$600 mln = mkt value of equity + $100 mln + $0 - $100 mln.

mkt value of equity = $600 mln.

Step 4: Estimate the share value

Share value = mkt value of equity / shares outstanding = $600/10 = $60 per share

7. You are tasked with valuing StellarBurst Technologies, which has an EBITDA of $50 million and 10 million shares outstanding. Additionally, StellarBurst Technologies has $50 million of cash and cash equivalent, $100 million of debt, and zero noncontrolling interests. You identify the following three comparable firms that have similar risk and growth as StellarBurst:

Comparable Firms	EV/EBITDA
CrimsonWave Solutions	8
NexusFusion Innovations	10
ZenithLabs Systems	9

What should be the estimated enterprise value of StellarBurst if you use the comparables/multiples approach?

Step 1: average EV/EBITDA of comparables = (8+10+9)/3=9

Step 2: estimated EV = 9 x target's EBITDA = 9 * 50 = $450 million

8. You are tasked with valuing StellarBurst Technologies, which has an EBITDA of $50 million and 10 million shares outstanding. Additionally, StellarBurst Technologies has $50 million of cash and cash equivalent, $100 million of debt , and zero noncontrolling interests. You identify the following three comparable firms that have similar risk and growth as StellarBurst:

Comparable Firms	EV/EBITDA
CrimsonWave Solutions	8
NexusFusion Innovations	10
ZenithLabs Systems	9

What should be the estimated total equity value of StellarBurst if you use the ccomparables/multiples approach?

Step 1: average EV/EBITDA of comparables = (8+10+9)/3=9

Step 2: estimated EV = 9 x target's EBITDA = 9 * 50 = $450 million

Step 3:

EV = Debt + Mkt value of equity + non-controlling interest - cash

estimate total equity value = EV + cash - Debt - non-controlling interest = $450 million + $50 million - $100 million - 0 =$400 million.

9. You are tasked with valuing StellarBurst Technologies, which has an EBITDA of $50 million and 10 million shares outstanding. Additionally, StellarBurst Technologies has $50 million of cash and cash equivalent, $100 million of debt (MV), and zero noncontrolling interests. You identify the following three comparable firms that have similar risk and growth as StellarBurst:

Comparable Firms	EV/EBITDA
CrimsonWave Solutions	8
NexusFusion Innovations	10
ZenithLabs Systems	9

What should be the estimated share value (equity value per share) of StellarBurst if you use the ccomparables/multiples approach?

Step 1: average EV/EBITDA of comparables = (8+10+9)/3=9

Step 2: estimated EV = 9 x target's EBITDA = 9 * 50 = $450 million

Step 3:

EV = Debt + Mkt value of equity + non-controlling interest - cash

estimate total equity value = EV + cash - Debt - non-controlling interest = $450 million + $50 million - $100 million - 0 =$400 million.

Step 4: equity value per share = total equity value / shares outstanding = $400 million /10 million = $40 per share

10. Infotech Ltd has earnings per share (EPS) of $1 per share. Table below provided selected financials of the comparable companies. Given these data, what is the estimated share value of Infotech Ltd. Stock?

Comparable Companies	Stock price	EPS
A	$20.00	$4.00

B	$30.00	$5.00
C	$40.00	$10.00

Step 1: calculate the P/E of each comparable:

A = 20/4=5

B= 30/5=6

C=40/10=4

Step 2: calculate the average P/E = (5+6+4)/3=5

Step 3: use the average P/E from step 2 x the target's EPS = estimate share value of the target = 5 * $1 = $5 per share.

11. Vexa Ltd has earnings per share (EPS) of $3 per share and 100 million shares outstanding. Table below provided selected financials of the comparable companies. Given these data, what is the estimated total equity value of Vexa Ltd. ?

Comparable Companies	Stock price	EPS
Zyxel	$20.00	$4.00
Quix	$30.00	$5.00
Xero	$40.00	$10.00

Step 1: calculate the P/E of each comparable:

A = 20/4=5

B= 30/5=6

C=40/10=4

Step 2: calculate the average P/E = (5+6+4)/3=5

Step 3: use the average P/E from step 2 x the target's EPS = estimate share value of the target = 5 * $3 = $15 per share.

Step 4: estimate the total equity value = $5 per share x shares outstanding = $15 * 100 =$1500 million.

12. You are provided with the following information about a firm:

- The number of shares outstanding is 20 million.
- The current share price is $5 per share.

- The total value of the firm's debt is $50 million.
- The cash and cash equivalents of the firm are $5 million.
- The firm does not have any non-controlling interests.
- The firm's EBITDA is $29 million.

What is the EV/EBITDA multiple of the firm?

Step 1: Mkt value of equity

Mkt value of equity is also called "mkt cap": = shares outstanding x share price = 20 mln x $5 = $100 mln.

Step 2: EV

EV = Mkt value of equity + debt + noncontrolling interest (also called minority interest) - cash & cash equivalents = $100 mln + $50 mln + 0 - $5 mln = $145 mln.

Step 2: EV/EBITDA

EV/EBITDA = $145 mln/$29 mln = 5

13. You are tasked with valuing Velocity Motors, which has an EBITDA of $30 million and 20 million shares outstanding. Additionally, Velocity Motors has $20 million of cash and cash equivalent, $10 million of debt , and zero noncontrolling interests. You identify the following three comparable firms that have similar risk and growth as Velocity Motors:

Comparable Firms	EV/EBITDA
EcoGear Ltd	4
SpeedCraft Ltd	5
RevMotors Corp.	6

Your company decides to acquire Velocity Motors' 100% shares outstanding. What should be the offer price per share if your company decides to pay 50% acquisition premium on top of the estimated share value of Velocity Motors based on the multiples approach?

Step 1: Average multiple of the comparable firm:

= (4+5+6)/3=5

Step 3: Estimate the enterprise value (EV)

= average multiple from step 1 x the target's EBITDA= 5 x $30 mln = $150 mln.

Step 4: Estimate the market value of equity or market cap

Note: EV = mkt value of equity + debt + noncontroling interest - cash & cash equivalent

$150 mln = mkt value of equity + $10 mln + $0 - $20 mln.

mkt value of equity = $160 mln.

Step 4: Estimate the share value

Share value = mkt value of equity / shares outstanding = $160/20 = $8 per share

Step 5: Estimate the offer price = $8 * (1+50%) = $12.

14. You are tasked with valuing StellarBurst Technologies, which has an EBITDA of $50 million and 10 million shares outstanding. Additionally, StellarBurst Technologies has $50 million of cash and cash equivalent, $100 million of debt , and zero noncontrolling interests. You identify the following three comparable firms that have similar risk and growth as StellarBurst:

Comparable Firms	EV	EBITDA	EBITDA growth rate
CrimsonWave Solutions	$160 mln	$20 mln	4%
NexusFusion Innovations	$200 mln	$20 mln	4%
ZenithLabs Systems	$270 mln	$30 mln	3%

StellarBurst's EBITDA is expected to growth at 3% per year. What should be the estimated total equity value of StellarBurst if you use the growth-adjusted multiple approach?

Step 1: Calculate the growth adjusted EV/EBITDA of each comparable company:

- CrimsonWave Solutions = 160/(20*4%) =2/%
- NexusFusion Innovations = 200/(20*4%) =2.5/%
- ZenithLabs Systems = 270/(30*3%) =3/%

Step 2: average EV/EBITDA of comparables = (2+2.5+3)/3=2.5/%

Step 3: estimated EV = $ x target's EBITDA* target's EBITDA growth rate = 2.5 * 50*3 = $375 million

avg of comp

Step 3:

EV = Debt + Mkt value of equity + non-controlling interest - cash

estimate total equity value = EV + cash - Debt - non-controlling interest = $375 million + $50 million - $100 million - 0 =$325 million.

Module 3 M&A valuation with DCF

1. You are tasked with valuing an unlevered target firm. The target firm is expected to generate an FCFF of $5 million one year from today. After that, the FCFF will grow at a constant rate of 4% per year. The acquirer's WACC is 8% and has $20 million of debt. The target's WACC is 12%. The target does not have any non-operating assets. What should be the value of the target's equity?

Step1: Find the target's WACC:

We should use the target's WACC to calculate the stand-alone value of the target.

Step2: Find the value of assets.

If the FCFFs are expected to grow at a constant rate, then we can use the growing perpetuity formula to calculate the value of the firm's assets:

PV = FCFF1/(WACC - g) = $5 mln / (12% - 4%) = $62.5 million.

This target firm is unlevered. So the total equity value = $62.5 million.

2. You are tasked with valuing a target firm. The target firm is expected to generate an FCFF of $5 million one year from today. After that, the FCFF will grow at a constant rate of 4% per year. The acquirer's WACC is 8%. The target's WACC is 12%. The target does not have any non-operating assets, and the market value of the target's debt is $12.5 million. What should be the target's share value (equity per share) if the target has 10 million shares outstanding?

Step1: Find the target's WACC:

We should use the target's WACC to calculate the stand-alone value of the target.

Step2: Find the value of the target's assets.

If the FCFFs are expected to grow at a constant rate, then we can use the growing perpetuity formula to calculate the value of the firm's assets:

PV = FCFF1/(WACC - g) = $5 mln / (12% - 4%) = $62.5 million.

The total equity value = total firm value - debt value = $62.5 million -$12.5 million =$50 million.

The target's share value = $50 million /10 million shares = $5 per share.

3. You are tasked with valuing a target firm. The target firm is expected to generate an FCFF of $5 million one year from today. After that, the FCFF will grow at a high growth rate of 10% per year for one year. After year 2, the FCFF will grow at constant rate of 5% per year. The target's WACC is 10%. The target does not have any non-operating assets. What should be the value of the target's assets ?

Step 1: Calculate FCFF2 = FCFF1 *(1+10%) = 5 * 1.1 = $5.5 million.

Step 2: Calculate FCFF3 = FCFF2 * (1+5%) = $5.775 million.

Step 3: Calculate Terminal Value at t=2 or TV2= FCFF3/(WACC-g) = 5.775/(10%-5%) = $115.5 million.

Step 4: Value of assets= PV of FCFF1 + PV of FCFF2 + PV of TV2 = 5/(1+10%) + 5.5/(1+10%)^2+115.5/(1+10%)^2= $104.55 million .

WACC (discount rate) to bring future cash flow back to present value

4. An all-equity firm's historical average Net Reinvestment Rate and After-tax Return on Capital are 20% and 10%, respectively. The firm's equity beta is 1, the market risk premium is 6%, and the risk-free rate is 3%. The firm is expected to have an FCFF of $30 million one year from today. You assume that the FCFFs of the firm will grow at a constant rate which is the same as the firm's long-term EBIT growth rate. The firm does not have any non-operating assets. What is the total value of the firm?

Step 1: Calculate the firm's long-term EBIT growth:

= After-tax ROC x Net investment Rate = 10% x 20% = 2%.

Step 2: Calculate the firm's WACC:

WACC= Cost of equity since the firm is an all-equity firm

re = rf + beta x MRP = 3% + 1x 6% = 9%.

Step 3: Calculate the firm's non-operating assets:

PV = FCFF1/(WACC - g) = $30 mln / (9% - 2%) = $428.57 mln.

Step 4: since the firm has no non-operating assets, the total firm value = $428.57 mln.

5. You are tasked with valuing a target firm. The target firm is expected to generate an FCFF of $2 million one year from today. After that, the FCFF will grow at a constant rate of 3% per year. The acquirer's WACC is 7%. The target's WACC is 11%. What should be the value of the target's assets?

Step1: Find the target's WACC:

We should use the target's WACC to calculate the stand-alone value of the target.

Step2: Find the value of operating assets.

If the FCFFs are expected to grow at a constant rate, then we can use the growing perpetuity formula to calculate the value of the firm's operating assets:

PV = FCFF1/(WACC - g) = $2 mln / (11% - 3%) = $25 million.

6. A firm's WACC is 9%. The firm is expected to have an FCFF of $6 million one year from today. You assume that the FCFFs of the firm will grow at a constant rate of 3%. The firm does not have any non-operating assets or noncontrolling interests. The firm has 10 million shares

outstanding. The market value of the firm's debt and other non-current liabilities is estimated to be $50 million. What is the estimated equity value per share?

Answer:

Step 1. Estimate total firm value

Since the firm's FCFFs are expected to grow at a constant rate g =3%.

$$V(operating\ assets) = \frac{FCFF_1}{WACC - g} = \frac{6}{9\% - 3\%} = \$100\ million$$

Note that the firm has no operating assets.

$Total\ firm\ value\ = \$100\ million$

Step 2. Estimate total equity value

$Total\ firm\ value\ = Equity + Debt + Other\ Noncurrent\ Liabilities$

$Equity\ = Total\ firm\ value - Debt - Other\ Noncurrent\ Liabilities = 100 - 50 = \$50\ million$

Step 3. Estimate share value

Share value = total equity value/shares outstanding =$50 million /10 million =$5.

7. Your firm is considering a potential target firm, CSH Inc. You are provided with CSH Inc's historical income statement and balance sheet for the current year (year 0), as well as its Pro-forma financial statements for the next year (year 1):

Income Statement (year 0 and year 1, in thousand dollars)		
	Current	**Pro Forma**
Year	**Year 0**	**Year 1**
Sales	$97,194	$113,557
Cost of Goods Sold		
Direct Material	($20,532)	($23,752)
Direct Labour	($23,784)	($28,333)
Gross Profit	$52,878	$61,472
Sales and Marketing	($16,037)	($20,440)
Administrative	($14,579)	($17,034)

EBITDA	$22,262	$23,999
Depreciation	($5,995)	($5,946)
EBIT	$16,267	$18,053
Net Interest Expenses	($6,600)	($6,600)
EBT	$9,667	$11,453
Income Tax (30%)	($2,900)	($3,436)
Net Income	$6,767	$8,017

Balance Sheet (Year 0 and Year 1, in thousand dollars)		
	Current	**Pro Forma**
Year	**Year 0**	**Year 1**
Assets		
Cash	$15,000	$15,000
Accounts Receivable	$8,966	$13,001
Inventories	$7,151	$8,374
Total Current Assets	$31,117	$36,375
Property, Plant, & Equipment	$139,565	$136,565
Total Assets	$170,682	$172,940
Liabilities		
Account Payable	$6,085	$7,313
Long-term Debt	$110,000	$110,000
Total Liabilities	$116,085	$117,313
Shareholder's Equity		
Total Shareholders' Equity	$54,597	$55,627
Total Liabilities and Equity	$170,682	$172,940

You are provided some additional information of the target firm (CSH Inc.):

- The expected capital expenditure in year 1 is $2946 (in thousand dollars).
- CSH Inc. targets to maintain a D/E ratio of 2 in long run.
- The company does not have any non-operating assets such as excess cash.
- The corporate tax rate is 30%.
- The company's pre-tax cost of debt of 5%.

- The risk-free rate is 3%.
- The expected market risk premium is 6%.
- The company's current equity beta is 1.
- The long-run after-tax return on capital is 10%.
- The net reinvestment rate is 20%.

A) What is the firm's WACC?

B) What is the firm's estimated FCFF in year 1?

C) What is the value of the firm's assets? D) Assumes that the market value of debt is the same as the book value. What is the estimated equity ?

<u>A) Calculate the WACC of the company</u>

- Step 1: cost of equity
  - $r_e = r_f$+beta*Market Risk Premium = 3%+1*6%=9%
- Step 2: calculate the weights of debt and equity
  - If D/E=2, then
    - W_D = D/(D+E)=2/(2+1) =2/3
    - W_E = E/(D+E)=1/(2+1)=1/3
- Step 3: calculate the WACC
  - WACC $=W_D*R_D*(1-T_C)+W_E*r_e$=2/3*5%*0.7+1/3*9%=5.33%

B) Calculate the expected FCFF in year 1

Note we should exclude non-operating cash in net working capital calculation (in this question, we are only provided with operating cash, so we don't need to do anything with it).

<u>Step 1: Change in NWC</u>

- NWC in year 1 = (Cash + A.R. + Inventories) – A.P. = 15,000 + 13,001+8,374 - 7,313 = 29,062.
- NWC in year 0 = (Cash + A.R. + Inventories) – A.P. = 15,000 + 8,966 +7,151 - 6,085 = 25,032.
- Change in NWC = NWC year 1 – NWC year 0 = 29,062 - 25,032 = $4,030.

Step 2: FCFF

- FCFF = EBIT*(1-Tc) + Depreciation & Amortisation – ΔNWC – Capital Expenditure
- FCFF = 18,053*(1-0.3) + 5946 – 4030 -2946 = 11,607

Step 3: the g

- g = net reinvestment rate x after-tax return on capital
- g= 20% x 10% = 2%

Step 4: Estimate the total standalone value of the firm's operating assets

- In fact, all the future cash flows will be growing at a constant rate g. Therefore the total standalone value of the firm's operating should be the PV of a growing perpetuity:
- Value of assets = 11,607/(5.33%- 2%)=$348,213 (Thousands)
- Note that there could be some rounding error (which is fine) in your final answer.

D)

Total firm value or the value of assets = $348,213

Equity value = total firm value minus debt value

Equity value = 348,213 (from part C) - $110,000 = $238,213 thousand.

Module 4 Other issues in valuation

1. NovusTech Innovations Inc. has three divisions: semi-conductors, telecommunications, and data processing. You are provided with the EBITDA of each division and the average EV/EBITDA of each division's comparable companies.

Given the data in the table, what is the estimated enterprise value of NovusTech Innovations Inc. using the "sum-of-parts" approach?

Division	EBITDA	EV/EBITDA multiple (comparable firms)
Semi-conductors	$100 million	4
Telecommunications	$80 million	5
Data processing	$120 million	5

Step 1: calculate the estimated EV of each division:

Semi-conductors division's EV = $100 million x 4 = $400 million

Telecommunications division's EV = $80 million x 5 = $400 million

Data processing division's EEV = $120million x 5 =600 million

Step 2: "sum-of-parts" enterprise value

Total EV = $400 m + $400 m + $600 m = $1400 million

2. NovusTech Innovations Inc. has three divisions: semi-conductors, telecommunications, and data processing. You are provided with the EBITDA of each division and the average EV/EBITDA of each division's comparable companies.

Given the data in the table, what is the estimated <u>total equity value</u> of NovusTech Innovations Inc. using the "sum-of-parts" approach? NovusTech Innovations Inc. has $400 million of debt, $200 million of cash, and $0 non-controlling interests.

Division	EBITDA	EV/EBITDA multiple (comparable firms)
Semi-conductors	$100 million	4
Telecommunications	$80 million	5
Data processing	$120 million	5

Step 1: calculate the estimated EV of each division:

Semi-conductors division's EV = $100 million x 4 = $400 million

Telecommunications division's EV = $80 million x 5 = $400 million

Data processing division's EEV = $120million x 5 =600 million

Step 2: "sum-of-parts" EV

Total EV = $400 m + $400 m + $600 m = $1400 million

Step 3: total value of equity

EV = Debt + Mkt Value of Equity + non-controlling interest - cash

$1400 m = $400 m + Mkt Value of Equity + 0 - $200 m

Estimated Mkt value of equity = $1400 m - $400 m - 0 +200 m = $1200 m

3. NovusTech Innovations Inc. has three divisions: semi-conductors, telecommunications, and data processing. You are provided with the EBITDA of each division and the average EV/EBITDA of each division's comparable companies.

Given the data in the table, what is the estimated <u>share value</u> (equity value per share) of NovusTech Innovations Inc. using the "sum-of-parts" approach? NovusTech Innovations Inc. has $400 million of debt, $200 million of cash, $0 non-controlling interests, and 50 million shares outstanding.

Division	EBITDA	EV/EBITDA multiple (comparable firms)
Semi-conductors	$100 million	4
Telecommunications	$80 million	5
Data processing	$120 million	5

Step 1: calculate the estimated EV of each division:

Semi-conductors division's EV = $100 million x 4 = $400 million

Telecommunications division's EV = $80 million x 5 = $400 million

Data processing division's EEV = $120million x 5 =600 million

Step 2: "sum-of-parts" EV

Total EV = $400 m + $400 m + $600 m = $1400 million

Estimated Mkt value of equity = $1400 m - $400 m - 0 +200 m = $1200 m

Step 3: total value of equity

EV = Debt + Mkt Value of Equity + non-controlling interest - cash

Step 4: share value

Share value = total equity value / # of shares outstanding = $1200 million/50 million = $24 per share

4. NovusTech Innovations Inc. has three divisions: semi-conductors, telecommunications, and data processing. You are provided with the EBITDA of each division and the average EV/EBITDA of each division's comparable companies.

Division	EBITDA	EV/EBITDA multiple (comparable firms)
Semi-conductors	$100 million	4
Telecommunications	$80 million	5
Data processing	$120 million	5

NovusTech Innovations Inc. has $200 million of cash, $0 non-controlling interests, and 50 million shares outstanding. Given the data in the table above, the estimated share value (equity value per share) of NovusTech Innovations Inc. using the "sum-of-parts" approach is $24 per share. What is total debt value of NovusTech Innovations Inc.?

Step 1: calculate the estimated EV of each division:

Semi-conductors division's EV = $100 million x 4 = $400 million

Telecommunications division's EV = $80 million x 5 = $400 million

Data processing division's EEV = $120million x 5 =600 million

Step 2: "sum-of-parts" EV

Total EV = $400 m + $400 m + $600 m = $1400 million

Step 3: total debt value

Mkt value of equity = $24 x shares outstanding = $24 * 50 million = $1200 million

EV = Debt + Mkt Value of Equity + non-controlling interest - cash

$1400 m = debt+ 1200 m + 0 - 200 m

debt = 1400 m - 1200 m+ 200m = $400 million

5. NovusTech Innovations Inc. has three divisions: semi-conductors, telecommunications, and data processing. You are provided with the EBITDA of each division and the average EV/EBITDA of each division's comparable companies.

Division	EBITDA	EV/EBITDA multiple (comparable firms)
Semi-conductors	$100 million	4
Telecommunications	$80 million	5
Data processing	$120 million	5

NovusTech Innovations Inc. has $300 million of cash, $0 non-controlling interests, and 50 million shares outstanding. Given the data in the table above, the estimated share value (equity value per share) of NovusTech Innovations Inc. using the "sum-of-parts" approach is $24 per share. What is total debt value of NovusTech Innovations Inc.?

Step 1: calculate the estimated EV of each division:

Semi-conductors division's EV = $100 million x 4 = $400 million

Telecommunications division's EV = $80 million x 5 = $400 million

Data processing division's EEV = $120million x 5 =600 million

Step 2: "sum-of-parts" EV

Total EV = $400 m + $400 m + $600 m = $1400 million

Step 3: total debt value

Mkt value of equity = $24 x shares outstanding = $24 * 50 million = $1200 million

EV = Debt + Mkt Value of Equity + non-controlling interest - cash

$1400 m = debt+ 1200 m + 0 - 300 m

debt = 1400 m - 1200 m+ 300m = $500 million

6. Your company plans to acquire 100% of Welling Inc.'s equity. You are given the following forecasts of the target firm:

- The FCFF one year from today ($FCFF_1$) will be $5 million.
- The FCFF will grow at a constant rate of 3% per year perpetually.
- The target's cost of debt is 6%.
- The target's long-term targeted D/E is 0.1.

Welling Inc. is a privately held firm. You find three listed companies that have similar business nature as Welling Inc. You run regressions to estimate the equity beta of the three comparable companies and get the following information:

Comparable companies	Equity Beta	D/E
X	1.53	1
Y	1.29	1.2
Z	1.25	0.8

The risk-free rate is 3%. The market risk premium is 6%. The corporate tax rate is 30%. The target firm has zero non-controlling interests and zero non-operating assets.
If we ignore the private company discount, what is the value of the target's assets?

- Step 1: Calculate the average unlevered beta of the three comparables:
    - X: $\beta_e = \beta_u \times \left[1 + \frac{D}{E} \times (1 - T_C)\right]$
        - $1.53 = \beta_u \times [1 + 1 \times (1 - 0.3)]$
        - $\beta_u = 0.9$
    - Y: $\beta_e = \beta_u \times \left[1 + \frac{D}{E} \times (1 - T_C)\right]$
        - $1.29 = \beta_u \times [1 + 1.2 \times (1 - 0.3)]$
        - $\beta_u = 0.7$
    - Z: $\beta_e = \beta_u \times \left[1 + \frac{D}{E} \times (1 - T_C)\right]$
        - $1.25 = \beta_u \times [1 + 0.8 \times (1 - 0.3)]$
        - $\beta_u = 0.8$
    - Average unlevered beta= (0.9+0.7+0.8)/3=0.8

- Step 2: WACC
    - We assume that the target's unlevered beta is the same as the average unlevered beta of the three comparables.
    - Target's levered beta $\beta_e = \beta_u \times \left[1 + \frac{D}{E} \times (1 - T_C)\right] = 0.8 \times [1 + 0.1 \times (1 - 0.3)] = 0.856$
    - Target's cost of equity $r_e = r_f + \beta_e \times MRP = 3\% + 0.856 \times 6\% = 8.136\%$
    - Target's WACC

- $W_d = \frac{D}{D+E} = \frac{0.1}{0.1+1} = 0.091$
- $W_e = \frac{E}{D+E} = \frac{1}{0.1+1} = 0.909$
- $WACC = W_d \times r_d \times (1 - T_C) + W_e \times r_e = 0.091 \times 6\% \times (1 - 0.3) + 0.909 \times 8.136\% = 7.78\%$

- Step 3: Target value
  - The FCFFs will grow at a constant rate, so they are a perpetuity.
  - Total firm value or the value of assets $= \frac{FCFF_1}{r-g} = \frac{5}{0.078-0.03} =$ $104.64 mln